

Dhruv Malhotra

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SKILLS

Programming/ML: Python, Java, Django, PyTorch, Pandas, TensorFlow, Flask, React.js, C/C++, Swift
Cloud/Tools: AWS (EC2, Lambda), Docker, Jenkins, Git, Postman, MySQL, PostgreSQL, Browserstack

WORK EXPERIENCE

Software Engineer in Test, Eagle Eye Networks

Austin TX | October 2024 – Present

- Built a fault-tolerant Gmail ingestion pipeline to batch-process 200+ emails/day, enrich alert data via API, and export to CSV; improved throughput 5× using concurrency, caching, and backoff logic
- Developed a high-concurrency API testing tool (Python + aiohttp) with session auth, dynamic CLI filtering, pagination, and Docker CI/CD integration for benchmarking Eagle Eye's core APIs
- Benchmarked core APIs (/alerts, /alertActions, /alertConditionRules, /notifications), automating P95/P99 latency, throughput, and error metrics to establish baselines and surface filter-driven bottlenecks
- Led notification parity testing between legacy v1 and v3 APIs for Aggregate Industries; built a parity matrix, identified latency gaps, and delivered a deprecation roadmap adopted by Customer Success
- Authored 1,000+ unit tests (pytest-django) and 300+ Postman API tests, achieving 100% coverage across core services and thoroughly validating authentication, role-based permissions, and complex end-to-end workflows
- Replaced PostgreSQL with SQLite for automated test runs, reducing runtime by 40%, and simulated 500+ virtual camera streams with concurrency modeling to enable realistic, large-scale REST API stress testing
- Developed Python automation to compare notification timing across enterprise monitoring systems, analyzing 4,600+ alerts on 10+ endpoints to reach 92% correlation accuracy and uncover 25% performance gains.

Software Engineer, Yunex Traffic

Austin TX | July 2023 – October 2024

- Developed real-time traffic management software in Python, integrating complex scheduling algorithms and database-backed state handling, which improved urban traffic flow efficiency 15% across multiple city networks
- Integrated AI/ML models with TensorFlow and Python for predictive traffic analysis, applying time-series forecasting and anomaly detection to cut peak-hour congestion 20% and reduce operational costs 30%
- Migrated legacy traffic management systems to AWS, redesigning services for cloud-native deployment and leveraging Docker + Kubernetes clusters to modernize infrastructure, improve scalability, and system reliability
- Migrated legacy systems to AWS, deploying with Docker + Kubernetes for scalable containerized environments
- Built a load testing framework simulating 3,000+ intersections, incorporating concurrency modeling and data-driven traffic scenarios to validate reliability and scalability of large-scale traffic management systems
- Implemented testing protocols using Pytest, ensuring a 98% success rate in bug resolution

Software Engineer, Givelify

Austin TX | May. 2020 – July 2023

- Collaborate cross-functionally with agile engineering, product, and quality teams to develop and deploy a recommendation engine using Python and Pytorch, elevating user retention by 25% and engagement by 20%
- Refined search capabilities using Python and React, benefitting 10k+ daily users by boosting efficiency by 15%
- Enhanced application performance using optimized data structures Python, improving speeds by 35%
- Integrated AI/ML tools with existing tech stack, contributing to innovation and platform scalability, as well as Geocoding services using Google API and LocationIQ, enhancing location-based search for 5k+ daily users
- Improved donation workflows by building new API endpoints and streamlining payment integrations, reducing transaction latency by 18% and enhancing reliability of high-volume transactions for thousands of daily users

Junior Software QA Developer, Nourtek

Dallas, TX | October 2019 – May 2020

- Led 500+ software tests using Junit, enhancing software reliability by 20%
- Piloted regression testing using Jira, cementing software stability across 25+ releases

Navy Corpsman, United States Navy

Okinawa, JP | July 2011 – July 2016

- Managed medical care for 4.5k servicemembers: checkups, immunizations, preventative care
- Attached to S-2(Security and Intelligence Division) ensuring security clearances of 3000 servicemembers

EDUCATION

Bachelor of Science in Computer Science

University of Texas at Dallas, Richardson, TX | Aug. 2018 - Dec 2021